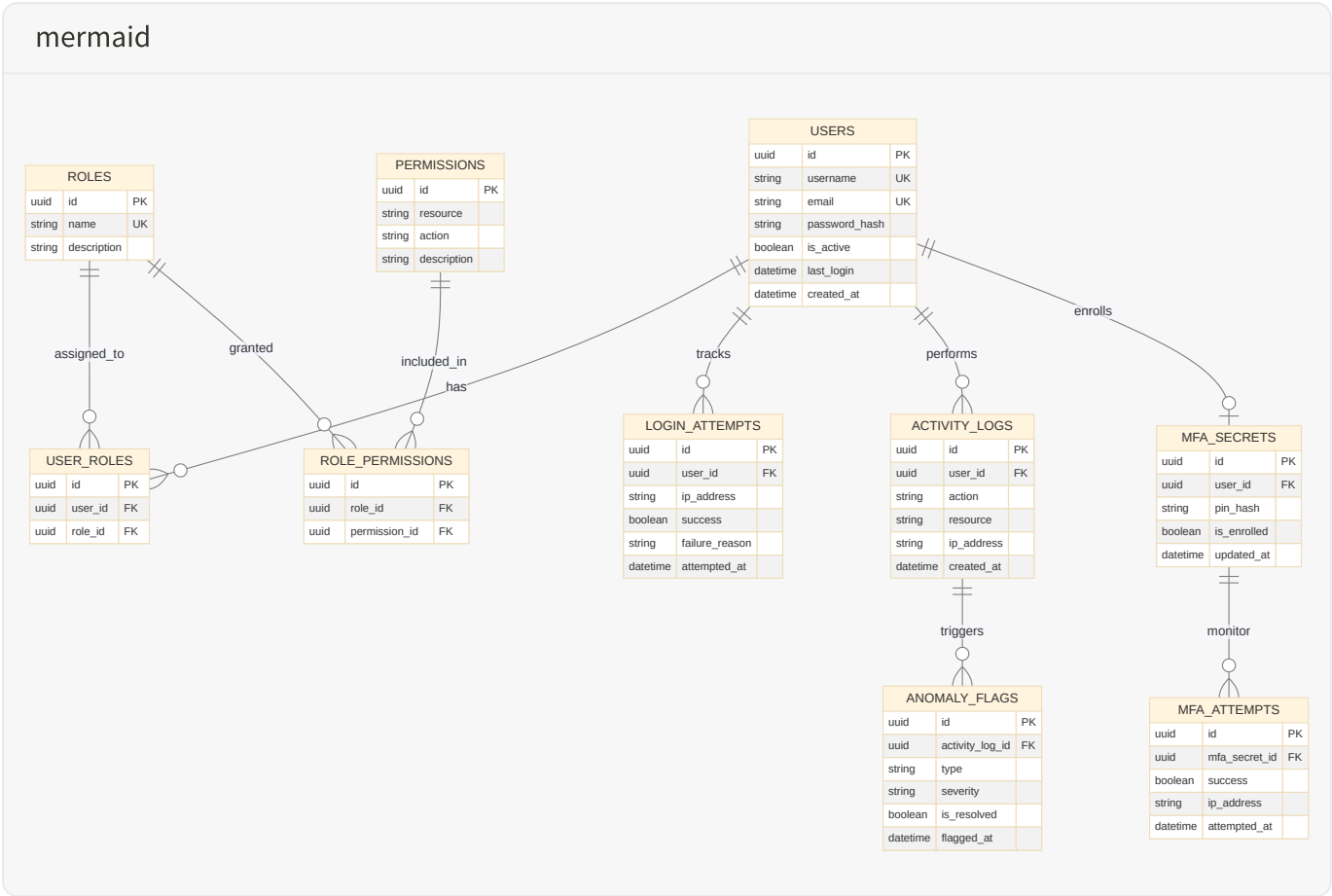


Master Entity Relationship Diagram (ERD)

IAM Security System (Sprints 1-4 + MFA)

The following diagram visualizes the unified architecture of the IAM Security System. It illustrates the relationships between core identity management, the multi-factor authentication (MFA) layer, and the defensive telemetry perimeter.



Key Architectural Patterns

- Identity Isolation:** Users and Roles are decoupled via a junction table to support many-to-many relationships.
- MFA Integration:** The MFA layer is an optional extension of the User entity, allowing for flexible enrollment.
- Forensic Persistence:** Activity and login logs maintain historical data even if users are deactivated or deleted (using `SET NULL` constraints in the schema).
- Anomaly Linkage:** Every anomaly flag is directly linked to the specific activity log entry that triggered it, providing immediate context for security analysts.